Hand-written data structures and where the pipeline uses them

Structure / technique	Used for	Cost
Dynamic array	event lists, CSR graph	push O(1) amortised
Hash table	NORAD-id lookup, grid cells, timeline	get/put O(1) expected
Binary heap	Top-K closest, greedy max-degree	push/pop O(log n)
Queue	BFS components, hop distance	O(1)
Stack	iterative DFS, undo in exact search	O(1)
AVL tree	time-ordered conjunction index	insert/find O(log n)
k-d tree	3-D radius query for candidate pairs	build O(n log n); query ~O(log n + k) typical
Uniform grid (hash)	cell lookup for candidate pairs	build O(n); query O(1) cells + k typical
Union-find (DSU)	cluster cross-check	near-constant per operation
Greedy / branch and bound	vertex-cover mitigation	greedy O(m log n); exact exponential worst case

Source: docs/dsa_coverage.md (use), standard textbook costs (cost column)